An Elementary Explicit Conditional Bound for a 2-Full Integer Followed by a 3-Full Integer

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Abstract

A positive integer is k-full if every prime dividing it occurs to exponent at least k. We consider the open problem of whether a 2-full integer n can have $n+1$ 3-full. Assuming Baker's *abc*-conjecture, we prove

$$n < 10^{16136778163}.$$

This proof was realized using assistance of GPT 5.6 Sol high.

1 Introduction

For $k \geq 2$, a positive integer m is *k-full* if

$$p \mid m \implies p^k \mid m$$

for every prime p. The 2-full integers are also called *powerful numbers*; see Golomb [3]. Erdős and Graham asked whether a 2-full integer n can be followed by a 3-full integer $n+1$ [2, p. 68]. The argument below does not settle existence, but makes the problem finite under one explicitly stated conjecture.

We write

$$\mathrm{rad}(m) = \prod_{p \mid m} p, \qquad \omega(m) = \#\{p : p \mid m\}.$$

All logarithms are natural.

2 Conjectural input

We assume the following conjecture of Baker [1].

Baker's explicit *abc*-conjecture. If a, b, c are pairwise coprime positive integers, $a + b = c$, and

$$N = \mathrm{rad}(abc) > 2, \qquad w = \omega(N),$$

then

$$c < \frac{6}{5} N \frac{(\log N)^w}{w!}. \tag{BABC}$$

This displayed inequality is the only conjectural input.

3 Elementary estimates

Lemma 1 (Factorial bounds)**.** *If N is a positive integer and $w = \omega(N)$, then*

$$N \geq (w+1)!, \tag{2.1}$$

and, for every integer $m \geq 1$,

$$m! > \sqrt{2\pi m}\left(\frac{m}{\mathrm{e}}\right)^m. \tag{2.2}$$

Proof. List the distinct prime divisors as $q_1 < \cdots < q_w$. Since they are distinct integers at least 2, $q_i \geq i+1$, and hence

$$N \geq \prod_{i=1}^{w} q_i \geq \prod_{i=1}^{w}(i+1) = (w+1)!.$$

Inequality (2.2) is the classical sharp lower Stirling inequality. It is, for example, an immediate consequence of Robbins's explicit remainder estimate [4], or of the elementary Stirling–Wallis proof. Equivalently,

$$\log(m!) > \left(m + \frac{1}{2}\right)\log m - m + \frac{1}{2}\log(2\pi).$$

Discarding the square-root term gives the weaker inequality $m! > (m/\mathrm{e})^m$ used at several points below. $\square$

For $X \geq \mathrm{e}^5$, define

$$A(X) = \log X - \log\log X - 1, \qquad G_0(X) = \frac{1+\log\log X}{\log X}, \qquad H(X) = \frac{1+\log A(X)}{A(X)}. \tag{2.3}$$

Lemma 2 (BABC multiplier)**.** *Let $N > 2$, put $X = \log N \geq \mathrm{e}^5$, and let $w = \omega(N)$. Then*

$$1 \leq w \leq \frac{X}{A(X)}. \tag{2.4}$$

Moreover $G_0(X) < H(X)$, and the following branch-specific estimates hold:

$$\begin{cases} \dfrac{X^w}{w!} < \exp\{XG_0(X)\}, & w < X/\log X, \\ \dfrac{X^w}{w!} < \dfrac{\exp\{XH(X)\}}{\sqrt{2\pi X/\log X}} < \dfrac{\exp\{XH(X)\}}{A(X)}, & w \geq X/\log X. \end{cases} \tag{2.5}$$

Proof. Since $N > 2$, it has a prime divisor and hence $w \geq 1$. At the left endpoint,

$$A(\mathrm{e}^5) = 4 - \log 5 > 2 > 1,$$

where $\log 5 < 2$ follows from $\mathrm{e}^2 > 1 + 2 + 2 = 5$. Moreover $A'(X) = (1 - 1/\log X)/X > 0$ for $X \geq \mathrm{e}^5$. Thus $1 < A(X) < \log X < X$ throughout the stated interval. From (2.1) and the weak form of Stirling,

$$X = \log N \geq \log(w!) > w(\log w - 1).$$

The function $t(\log t - 1)$ is increasing for $t > 1$. If $w > X/A(X)$, then

$$w(\log w - 1) > \frac{X}{A(X)}\left(\log\frac{X}{A(X)} - 1\right) > X.$$

The last inequality is equivalent to $\log A(X) < \log\log X$, true because $A(X) < \log X$. This contradiction proves (2.4).

For fixed X, put

$$g_X(t) = t\left(1 + \log\frac{X}{t}\right).$$

It is increasing for $0 < t < X$, since $g'_X(t) = \log(X/t) > 0$. If $w < X/\log X$, weak Stirling therefore gives

$$\log\frac{X^w}{w!} < g_X(w) < g_X\left(\frac{X}{\log X}\right) = XG_0(X).$$

If $w \geq X/\log X$, sharp Stirling, (2.4), and the same monotonicity give

$$\begin{aligned}\log\frac{X^w}{w!} &< g_X(w) - \frac{1}{2}\log(2\pi w)\\ &\leq g_X\left(\frac{X}{A(X)}\right) - \frac{1}{2}\log\frac{2\pi X}{\log X}\\ &= XH(X) - \frac{1}{2}\log\frac{2\pi X}{\log X}.\end{aligned}$$

This direction is important: the lower bound $w \geq X/\log X$, valid only in this branch, is what permits the replacement of $\sqrt{2\pi w}$ by $\sqrt{2\pi X/\log X}$ in the denominator.

It remains to justify the final, deliberately coarser denominator in (2.5). Since $A(X) < \log X$, it is enough that $2\pi X > (\log X)^3$. The exponential series gives $\mathrm{e}^5 > 100$, so $2\pi\mathrm{e}^5 > 200 > 125$; thus the inequality holds at $X = \mathrm{e}^5$. Also $X/(\log X)^3$ is increasing for $\log X > 3$. Hence

$$\sqrt{\frac{2\pi X}{\log X}} > A(X) \qquad (X \geq \mathrm{e}^5).$$

Finally, $F(t) = (1 + \log t)/t$ is strictly decreasing for $t > 1$, because $F'(t) = -\log(t)/t^2 < 0$. Thus

$$G_0(X) = F(\log X) < F(A(X)) = H(X).$$

□

Lemma 3 (Monotonicity)**.** *On* $[\mathrm{e}^5, \infty)$*,* A *is strictly increasing,* G_0 *and* H *are strictly decreasing, and*

$$P_0(X) = X(1 + G_0(X)), \qquad Q(X) = X(1 + H(X)) - \log A(X) \tag{2.6}$$

are strictly increasing. On any interval where $G_0(X) < 1/5$*, the function* $X(1/5 - G_0(X))$ *is also strictly increasing.*

Proof. Write $L = \log X$. Direct differentiation gives

$$A'(X) = \frac{1 - 1/L}{X} > 0, \qquad \frac{d}{dt}\frac{1 + \log t}{t} = -\frac{\log t}{t^2} < 0.$$

This proves the first three assertions. Furthermore

$$P_0'(X) = 1 + \frac{1 + \log L}{L} - \frac{\log L}{L^2} > 0.$$

If $P(X) = X(1 + H(X))$, then

$$P'(X) = 1 + H(X) - \frac{\log A(X)}{A(X)^2}\left(1 - \frac{1}{L}\right) > 1 - \frac{1}{2\mathrm{e}},$$

using $\max_{t\ge 1}\log(t)/t^2 = 1/(2\mathrm{e})$. Consequently

$$Q'(X) = P'(X) - \frac{1-1/L}{XA(X)} > 1 - \frac{1}{2\mathrm{e}} - \frac{1}{100} > 0,$$

where $\mathrm{e}^5 > 100$ and $A > 1$ justify the last error bound. Finally, where $G_0 < 1/5$, both positive factors in $X(1/5 - G_0(X))$ are strictly increasing. $\square$

When $0 < X < \mathrm{e}^5$, we shall use only the exponential-series estimate

$$\frac{X^w}{w!} < \sum_{j=0}^{\infty} \frac{X^j}{j!} = \mathrm{e}^X. \tag{2.7}$$

4 The explicit bound

Theorem 1. *Assume Baker's explicit abc-conjecture. If n is a positive integer, n is 2-full, and $n+1$ is 3-full, then*

$$\boxed{n < 10^{16136778163}}.$$

Proof. Apply (BABC) to $a = n$, $b = 1$, and $c = n+1$, and put $N = \mathrm{rad}(n(n+1))$. Coprimality and the fullness assumptions give

$$N = \mathrm{rad}(n)\,\mathrm{rad}(n+1) \le n^{1/2}c^{1/3} < c^{5/6}.$$

Thus

$$N^{6/5} < c. \tag{3.1}$$

The case $N \le 2$ is impossible. Indeed, $n(n+1)$ is even, so $N = 2$. The odd member of $n, n+1$ then has no prime divisor and equals 1; hence $n = 1$, but $n+1 = 2$ is not 3-full. We may assume $N > 2$.

Set

$$X_0 = 30963587374.$$

The outward-rounded certificate in Appendix A proves

$$\begin{aligned} 24.15607775020 < \log X_0 < 24.15607775021,\\ 19.97154173509 < A(X_0) < 19.97154173510,\\ 0.17322911684 < G_0(X_0) < 0.17322911685 < \frac{1}{5},\\ 0.1999999999993 < H(X_0) < 0.1999999999994 < \frac{1}{5}, \end{aligned} \tag{3.2}$$

and the two endpoint comparisons

$$X_0\left(\frac{1}{5} - G_0(X_0)\right) > \log\frac{6}{5}, \tag{3.3}$$

$$\frac{\log(6/5) + X_0(1 + H(X_0)) - \log A(X_0)}{\log 10} < 16136778163. \tag{3.4}$$

Put $X = \log N$. Suppose $X \ge X_0$. If $w < X/\log X$, then (BABC), (2.5), and (3.1) imply

$$\mathrm{e}^{X/5} < \frac{6}{5}\mathrm{e}^{XG_0(X)},$$

or $X(1/5 - G_0(X)) < \log(6/5)$. This contradicts Lemma 3 and (3.3). If $w \ge X/\log X$, the second branch of (2.5) instead gives

$$\mathrm{e}^{X/5} < \frac{6}{5A(X)}\mathrm{e}^{XH(X)} < \mathrm{e}^{X/5},$$

where $H(X) \le H(X_0) < 1/5$ and $A(X) \ge A(X_0) > 6/5$. This is again impossible. Thus

$$X < X_0. \tag{3.5}$$

The sharp square-root factor is essential in the large-w branch: without it, $X_0(1/5 - H(X_0)) \approx 0.0202$ is smaller than $\log(6/5) \approx 0.1823$. The valid branch-specific lower bound on $\sqrt{2\pi w}$ supplies much more than the factor $A(X) > 6/5$ used above.

Now suppose $\mathrm{e}^5 \le X < X_0$. In the small-w branch, (BABC), (2.5), monotonicity, and the certificate give

$$\log c < \log\frac{6}{5} + P_0(X) < \log\frac{6}{5} + P_0(X_0) < 16136778163\log 10.$$

In the large-w branch they give

$$\log c < \log\frac{6}{5} + Q(X) < \log\frac{6}{5} + Q(X_0) < 16136778163\log 10$$

by (3.4). If $0 < X < \mathrm{e}^5$, equations (BABC) and (2.7) give

$$\log c < \log\frac{6}{5} + 2X < \log\frac{6}{5} + 2\mathrm{e}^5 < 16136778163\log 10.$$

The two remaining endpoint comparisons are also certified in the appendix. Thus $c < 10^{16136778163}$ in every case, and $n < c$ completes the proof. □

Remark 1. *The same radical calculation gives the identical bound when n is 3-full and $n+1$ is 2-full. This is only a boundedness result: it neither produces a solution nor excludes all solutions below the bound.*

Remark 2. *The ordinary abc-conjecture gives finiteness but no numerical bound because its constant is unspecified. Here (BABC) is the only conjectural statement. The larger exponent is the cost of replacing explicit prime-counting estimates by the elementary factorial bound (2.1).*

A Interval-arithmetic certificate

The following SageMath script uses 256-bit outward-rounded real interval arithmetic. All integer and decimal inputs are exact before conversion to intervals, and every assertion evaluates to `True`. It certifies (3.2)–(3.4), both final branch estimates, and the small-X comparison. This Sage certificate independently mirrors the kernel-checked Lean certificates in extttErdos366/Numerics.lean; the Lean proof itself does not trust Sage or floating-point computation.

```
R = RealIntervalField(256)
Q = QQ

def inside(z, lo, hi):
    return z.lower() > Q(lo) and z.upper() < Q(hi)

X0 = R(Q(30963587374))
```

```
L  = X0.log()
A  = L - L.log() - 1
G0 = (1 + L.log()) / L
H  = (1 + A.log()) / A

assert inside(L,  "24.15607775020", "24.15607775021")
assert inside(A,  "19.97154173509", "19.97154173510")
assert inside(G0, "0.17322911684", "0.17322911685")
assert inside(H,  "0.1999999999993", "0.1999999999994")
assert G0.upper() < Q(1)/Q(5)
assert H.upper() < Q(1)/Q(5)
assert A.lower() > Q(6)/Q(5)

log65 = (R(6)/R(5)).log()
small_contradiction = X0*(R(1)/5-G0)
assert small_contradiction.lower() > log65.upper()

E_small = (log65 + X0*(1+G0)) / R(10).log()
E_large = (log65 + X0*(1+H) - A.log()) / R(10).log()
assert E_small.upper() < Q(16136778163)
assert E_large.upper() < Q(16136778163)

assert (log65 + 2*R(5).exp()).upper() \
       < (R(16136778163)*R(10).log()).lower()

# Diagnostic: coarse Stirling alone misses the 6/5 margin in
# the large branch, whereas the sharp square-root term suffices.
coarse = X0*(R(1)/5-H)
sharp = coarse + (2*R.pi()*X0/L).log()/2
assert coarse.upper() < log65.lower()
assert sharp.lower() > log65.upper()
assert (2*R.pi()*X0/L).sqrt().lower() > A.upper()
```

References

[1] A. Baker, *Experiments on the abc-conjecture*, Publ. Math. Debrecen **65** (2004), 253–260.

[2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique, no. 28, L'Enseignement Mathématique, Geneva, 1980.

[3] S. W. Golomb, *Powerful numbers*, Amer. Math. Monthly **77** (1970), 848–855.

[4] H. Robbins, *A remark on Stirling's formula*, Amer. Math. Monthly **62** (1955), 26–29.